Pilot: the 2020 Labor Day Fires (Holiday Farm Fire)

A small, cheap, end-to-end dry run of the entire wall-to-wall pipeline on a single well-measured disturbance, before committing to the full 0.73 TB/year run. Same steps as Wall_to_Wall_Maps.ipynb: AOI, export, containerized VM inference, mosaic; plus validation steps that test the science, not just the plumbing.

The event

The Holiday Farm Fire (McKenzie River corridor, OR), from MTBS: 174,390 acres, ignited 2020-09-08, 100% inside the forest AOI, spanning Lane and Linn counties and two ecoregions. ~70,600 ha is about 0.25% of the full job: pennies of storage, minutes of VM time.

Why this event

It gives independent ground truth on all three things the full pipeline claims to measure:

- Area: MTBS burn-severity strata say how much *actually* changed land-cover class.
- Date: a known ignition date (2020-09-08) to test CCDC (and alternatives) break-dating against.
- Carbon: dense Douglas-fir / western hemlock, a large real signal.

The central falsifiable prediction

A low-severity underburn leaves forest standing (still `Trees`); only stand-replacing fire flips the class. So the change map should recover roughly the moderate+high-severity area (~63% of the perimeter), not the whole fire. Detecting the full 174,390 ac would mean the map is *over*-detecting, which is exactly the classifier-noise bias the full study has to control. This turns "looks about right" into a number.

> Ordering caveat (same as the full run). The 2019-vs-2025 map difference is a candidate mask only; two independently classified maps disagree partly from model error. CCDC is the arbiter that confirms *and* dates real change. Carbon is attributed only to confirmed change.

Setup

In [1]:

```
%load_ext autoreload
%autoreload 2

# ── Load libraries and functions ──────────────────────────────────────────────
import os

import ee
import geemap
import pandas as pd

from Constants import MODEL_DIR, DATA_DIR, FIA_DATA_DIR, IPCC_CARBON_FRACTION, IPCC_CARBON_FRACTION_HI, CCDC_OUTPUTS_DIR
from Glance_Class_Definitions import class1_dict
from Map_Export import export_embeddings_to_gcs, get_embedding_image, wait_for_tasks
from Fire_Utils import (
    get_fire, fire_ignition_date, get_severity_image, severity_area_table,
    stand_changing_mask, SEVERITY_NAMES, STAND_CHANGING_SEVERITY,
)

# ── Initiate GEE Project ──────────────────────────────────────────────────────
ee.Initialize(project='turnkey-lacing-391919')

# ── Pilot variables ───────────────────────────────────────────────────────────
fire_name = 'HOLIDAY FARM'
map_years = [2019, 2025]     # bracket the 2020 fire; 2019 is pre-fire, 2025 post-fire
fire_year = 2020             # MTBS severity + ignition year
gcs_bucket = 'turnkey-lacing-391919-lulc'
gcs_prefix = 'pilot/embeddings'
map_dir = os.path.join('Data', 'Maps', 'pilot')
export_embeddings = False
download_predicted_tiles = False
rerun_CCDC = True
```

Step 1: Area of interest (the fire perimeter)

The pilot AOI is simply the MTBS fire perimeter, in place of the forest ecoregions used by the full run. Everything downstream (export, mask, inference, mosaic) is identical. The severity table below is the reference truth Step 6 validates against. Note the area is computed with `ee.Image.pixelArea()`, *not* a pixel count: the MTBS severity mosaic is served in EPSG:3338, so at this latitude a naive count would overstate area by ~1/cos(44°) ≈ 1.39.

In [2]:

```
# ── Fetch the fire perimeter + its reference severity table ───────────────────
fire = get_fire(fire_name)
fire_geometry = fire.geometry()
ignition = fire_ignition_date(fire)
print(f'{fire_name}: {fire.get("BurnBndAc").getInfo():,} acres, ignited {ignition}')

# The full-run export masks non-forest to nodata; here the whole perimeter is the AOI, so a
# constant mask (=1) keeps export_embeddings_to_gcs's signature identical with nothing dropped.
fire_mask = ee.Image.constant(1).clip(fire_geometry)

severity_table = severity_area_table(fire, fire_year)
print('\nMTBS burn severity within the perimeter (true ground area):')
display(severity_table.round(1))

detectable_ha = severity_table.loc[['moderate', 'high'], 'hectares'].sum()
total_ha = severity_table.loc['Total', 'hectares']
print(f'Expected detectable change (moderate+high): {detectable_ha:,.0f} ha '
      f'({detectable_ha / total_ha:.1%} of the perimeter)')
print('  -> the change map should recover ~this, NOT the full perimeter.')
```

```
HOLIDAY FARM: 174,390 acres, ignited 2020-09-08

MTBS burn severity within the perimeter (true ground area):
```

	hectares	fraction
unburned/low	6348.1	0.1
low	18927.2	0.3
moderate	20782.0	0.3
high	23431.6	0.3
increased greenness	569.8	0.0
non-mapping	239.5	0.0
Total	70298.2	1.0

```
Expected detectable change (moderate+high): 44,214 ha (62.9% of the perimeter)
  -> the change map should recover ~this, NOT the full perimeter.
```

In [3]:

```
# ── Visualize the fire: perimeter, severity, and the stand-changing mask ──────
sev_vis = {'min': 1, 'max': 6,
           'palette': ['#7fbf7f', '#ffff99', '#ffbf00', '#d7191c', '#1a9641', '#bdbdbd']}
m = geemap.Map(center=fire_geometry.centroid(maxError=100).coordinates().getInfo()[::-1], zoom=10)
m.addLayer(get_severity_image(fire_year).clip(fire_geometry), sev_vis, 'MTBS severity')
m.addLayer(stand_changing_mask(fire_year).clip(fire_geometry), {'palette': ['#d7191c']},
           'Stand-changing (moderate+high)')
m.addLayer(ee.Image().byte().paint(ee.FeatureCollection([fire]), 0, 2),
           {'palette': ['#000000']}, 'Perimeter')
m
```

Out[3]:

```
Map(center=[44.15068388319812, -122.47767176333126], controls=(WidgetControl(options=['position', 'transparent…
```

Step 2: Export AlphaEarth embeddings to Cloud Storage

Same `export_embeddings_to_gcs` as the full run, one task per map year (2019 pre-fire, 2025 post-fire), clipped to the fire perimeter. Band order is pinned to `A00` ... `A63`, CRS is EPSG:5070 for true 10 m pixels. At ~70,600 ha this is ~1.8 GB/year, trivial. Set `gcs_bucket` first.

In [4]:

```
# ── Submit one export task per map year ───────────────────────────────────────
if export_embeddings: #Only export embeddings if this is true since exporting to store is not a free decision
    assert gcs_bucket, 'Set gcs_bucket in the Setup cell before exporting.'

    tasks = []
    for year in map_years:
        task = export_embeddings_to_gcs(
            year, fire_geometry, fire_mask, bucket=gcs_bucket, prefix=gcs_prefix,
            description=f'pilot_holidayfarm_embeddings_{year}')
        tasks.append(task)
        print(f'submitted {year}: task {task.id} -> gs://{gcs_bucket}/{gcs_prefix}/{year}/')
else:
    print("Did not reexport new AlphaEarth embeddings to Cloud Storage")
```

```
Did not reexport new AlphaEarth embeddings to Cloud Storage
```

In [5]:

```
# ── Poll until every export reaches a terminal state ──────────────────────────
# Safe to interrupt and re-run: the tasks keep running server-side regardless.
if export_embeddings:
    final_states = wait_for_tasks(tasks, poll_seconds=30)
    print()
    for description, state in final_states.items():
        print(f'{description}: {state}')
```

Step 3: Containerized inference on a Cloud VM (runs outside this notebook)

Identical to the full run; the pilot exists partly to rehearse exactly this. The tuned XGBoost model is shipped as a Docker image (`Dockerfile`, `requirements-vm.txt`, `vm_predict.py`) and run on a temporary Compute Engine VM in the bucket's region. See Wall_to_Wall_Maps.ipynb Step 3 for the full write-up. In brief:

```
# Build + push (once); REGION/PROJECT/REPO are yours:
docker build -t REGION-docker.pkg.dev/PROJECT/REPO/lulc-inference:v1 .
docker push  REGION-docker.pkg.dev/PROJECT/REPO/lulc-inference:v1

# On a temporary e2-standard-4 VM in the bucket's region, per map year:
docker run --rm REGION-docker.pkg.dev/PROJECT/REPO/lulc-inference:v1 \
    --bucket BUCKET --embeddings-prefix pilot/embeddings/2019 \
    --predictions-prefix pilot/predictions/2019 --confidence-prefix pilot/confidence/2019
# ...repeat for 2025, then DELETE THE VM.
```

`vm_predict.py` writes two rasters per tile: `predictions/` (uint8 class 1-7) and `confidence/` (uint8 max-probability). The confidence layer is what the change step uses to separate real change from classifier noise; exporting it now avoids re-running inference later.

Model note (applies to the full run too)

The Level-1 model's `classes_` is `[1, 3, 4, 5, 6, 7]`. GLanCE's North America training data contains no Ice/Snow (class 2), so the model cannot predict it. Irrelevant for this low-elevation fire, but the full WA+OR AOI contains glaciated peaks (Rainier, Hood, Baker, Sisters) whose permanent ice will be misassigned. Flagged for the full run.

This is the one step not driven from the notebook; resume below once `pilot/predictions/` and `pilot/confidence/` are populated.

Step 4: Mosaic + clip (local)

In [6]:

```
# ── Download predicted tiles and mosaic into one raster per map year ──────────
# Requires: gsutil (or google-cloud-storage) + rasterio. Run after Step 3 populates
# gs://{bucket}/pilot/predictions/{year}/.
import glob
import subprocess

import numpy as np
import rasterio
from rasterio.merge import merge

os.makedirs(map_dir, exist_ok=True)
if download_predicted_tiles:
    class_maps = {}
    for year in map_years:
        local_dir = os.path.join(map_dir, f'predictions_{year}')
        os.makedirs(local_dir, exist_ok=True)
        subprocess.run(
            ['gsutil', '-m', 'cp', f'gs://{gcs_bucket}/pilot/predictions/{year}/*.tif', local_dir],
            check=True)

        tiles = [rasterio.open(f) for f in sorted(glob.glob(os.path.join(local_dir, '*.tif')))]
        mosaic, transform = merge(tiles)
        profile = tiles[0].profile
        profile.update(height=mosaic.shape[1], width=mosaic.shape[2], transform=transform)
        for t in tiles:
            t.close()

        out_path = os.path.join(map_dir, f'LULC_Class1_{fire_name.title().replace(" ", "_")}_{year}.tif')
        with rasterio.open(out_path, 'w', **profile) as dst:
            dst.write(mosaic)
        class_maps[year] = out_path
        print(f'{year}: mosaicked {len(tiles)} tiles -> {out_path}  {mosaic.shape}')
else:
    print("Did not redownload predicted tiles and mosiac them")
```

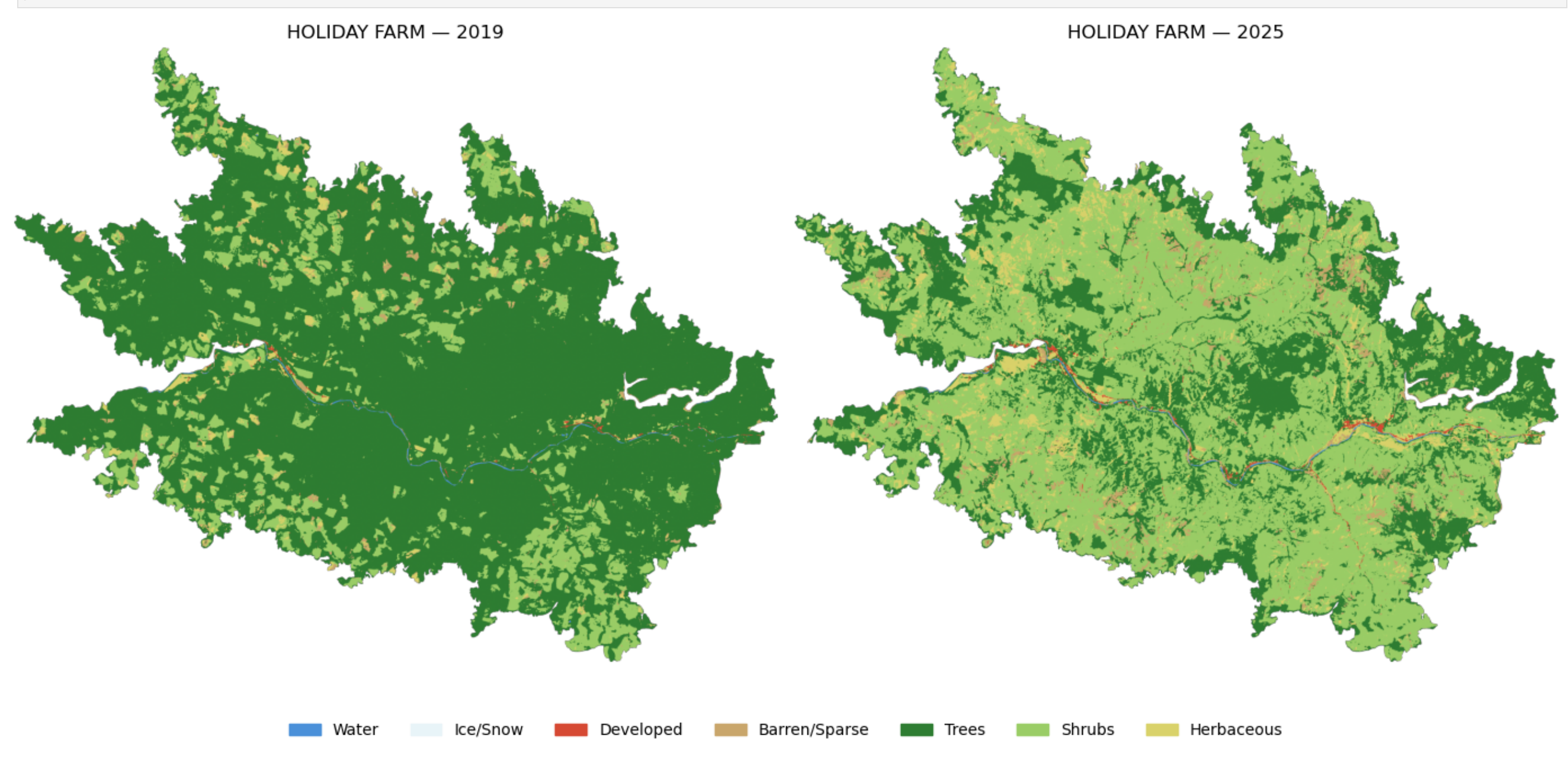
```
Did not redownload predicted tiles and mosiac them
```

Step 5: Visualize the before / after maps

The 2019 map should be almost entirely `Trees`. The 2025 map should show a burn scar over the moderate+high-severity footprint: Trees converted to Barren/Sparse, Herbaceous, or Shrubs. Anything else (Trees converted to Water/Developed) is a bug, not a fire.

In [7]:

```
# ── Plot both years side by side with a categorical colormap ──────────────────
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap, BoundaryNorm

# ── Point at the mosaics on disk, so this cell runs standalone ────────────────
# (Step 4 already wrote them; no need to re-download or re-mosaic.)
class_maps = {year: os.path.join(map_dir, f'LULC_Class1_{fire_name.title().replace(" ", "_")}_{year}.tif')
              for year in map_years}
for _y, _p in class_maps.items():
    assert os.path.exists(_p), f'Missing {_p} -- run Step 4 (mosaic) first.'

# class1_dict keys 1-7; NODATA is 255. Colours keyed to the GLanCE Level-1 classes.
class_colors = {1: '#4a90d9', 2: '#e8f4f8', 3: '#d64933', 4: '#c9a66b',
                5: '#2e7d32', 6: '#9acd66', 7: '#d9d26a'}
cmap = ListedColormap([class_colors[k] for k in sorted(class_colors)])
norm = BoundaryNorm([0.5 + k for k in range(0, 8)], cmap.N)

fig, axes = plt.subplots(1, len(map_years), figsize=(7 * len(map_years), 7))
for ax, year in zip(np.atleast_1d(axes), map_years):
    with rasterio.open(class_maps[year]) as src:
        arr = src.read(1).astype(float)
    arr[arr == 255] = np.nan
    ax.imshow(arr, cmap=cmap, norm=norm)
    ax.set_title(f'{fire_name} — {year}')
    ax.axis('off')
handles = [plt.Rectangle((0, 0), 1, 1, color=class_colors[k]) for k in sorted(class1_dict)]
fig.legend(handles, [class1_dict[k] for k in sorted(class1_dict)],
           loc='lower center', ncol=7, frameon=False)
plt.tight_layout()
plt.show()
```



HOLIDAY FARM — 2019

HOLIDAY FARM — 2025

Step 6: Validate against MTBS

The pass/fail checks. The headline one: detected Trees to non-Trees area should track the moderate+high-severity area (~63% of the perimeter), not the whole fire.

In [8]:

```
# ── Change map: pixels that went Trees (5) in 2019 -> non-Trees in 2025 ───────
with rasterio.open(class_maps[2019]) as s19, rasterio.open(class_maps[2025]) as s25:
    a19, a25 = s19.read(1), s25.read(1)
    pix_m2 = abs(s19.transform.a * s19.transform.e)  # true pixel area (EPSG:5070 -> ~100 m2)
    transform, crs = s19.transform, s19.crs          # kept for the county split below

valid = (a19 != 255) & (a25 != 255)
tree_loss = valid & (a19 == 5) & (a25 != 5)
detected_ha = tree_loss.sum() * pix_m2 / 1e4

detectable_ha = severity_table.loc[['moderate', 'high'], 'hectares'].sum()
high_ha = severity_table.loc['high', 'hectares']
print(f'Detected Trees->non-Trees : {detected_ha:>10,.0f} ha')
print(f'MTBS high severity        : {high_ha:>10,.0f} ha')
print(f'MTBS moderate+high        : {detectable_ha:>10,.0f} ha   <- detection should land near here')
print(f'MTBS full perimeter       : {severity_table.loc["Total","hectares"]:>10,.0f} ha')
print(f'\nRatio detected / (mod+high): {detected_ha / detectable_ha:.2f}   (want roughly 0.7-1.3)')

# ── What did Trees convert to? (should be Barren/Sparse, Herbaceous, or Shrubs) ──────────
to_classes = a25[tree_loss]
print('\nTrees converted to:')
for c in np.unique(to_classes):
    print(f'  {class1_dict.get(int(c), c):<16s} {(to_classes == c).sum() * pix_m2 / 1e4:>9,.0f} ha')
print('  (Water/Developed here might indicate a pipeline bug, not fire. Or could indicate changes during other times)')
```

```
Detected Trees->non-Trees :     36,167 ha
MTBS high severity        :     23,432 ha
MTBS moderate+high        :     44,214 ha   <- detection should land near here
MTBS full perimeter       :     70,298 ha

Ratio detected / (mod+high): 0.82   (want roughly 0.7-1.3)

Trees converted to:
  Water                   16 ha
  Developed              202 ha
  Barren/Sparse        1,899 ha
  Shrubs              31,396 ha
  Herbaceous           2,655 ha
  (Water/Developed here might indicate a pipeline bug, not fire. Or could indicate changes during other times)
```

Step 7: Date the change with CCDC

The area test (Step 6) shows *how much* changed; CCDC shows *when*. It runs on Landsat surface reflectance, not AlphaEarth: CCDC fits a seasonal harmonic model to dense intra-annual observations and flags a break when the pattern shifts persistently, and AlphaEarth is only one vector per year.

Design

A set number of random points from the classifier's Trees to non-Trees change map (Step 6), plus a stratified addition of Trees to Water and Trees to Developed points, searched for any break across 2019-2025. The random points alone almost never include Water/Developed by chance; those transitions are under 1% of the total change area combined, so they're added on top, clearly tagged (`sample_type`), rather than folded into the "random" draw (which stays a genuine unbiased sample).

Comparison to other methods

CCDC was originally ran with minObservations set to 6, but that caused the algorithm to predict transitions months after their timeseries visually showed transitions around when the fire was. Changing minObservations to 4 improved this greatly. I also tested another land use change algorithm (COLD), which did improve predictions based on visually comparing time series of NDVI and NBR (burn index). The CCDC is able to be run on GEE engine for free, but COLD needs to run on a virtual machine, costing money. Because of this, CCDC with minObservations of 4 is likely going to be used for the wider case study. For a more detailed comparison of the methods see the COLD_vs_CCDC_Comparison.pdf file.

CCDC is time expensive: over the whole perimeter it exceeds Earth Engine's interactive memory limit (confirmed the hard way). So the pilot buffers each sampled point to a tiny region and runs CCDC there, one point at a time.

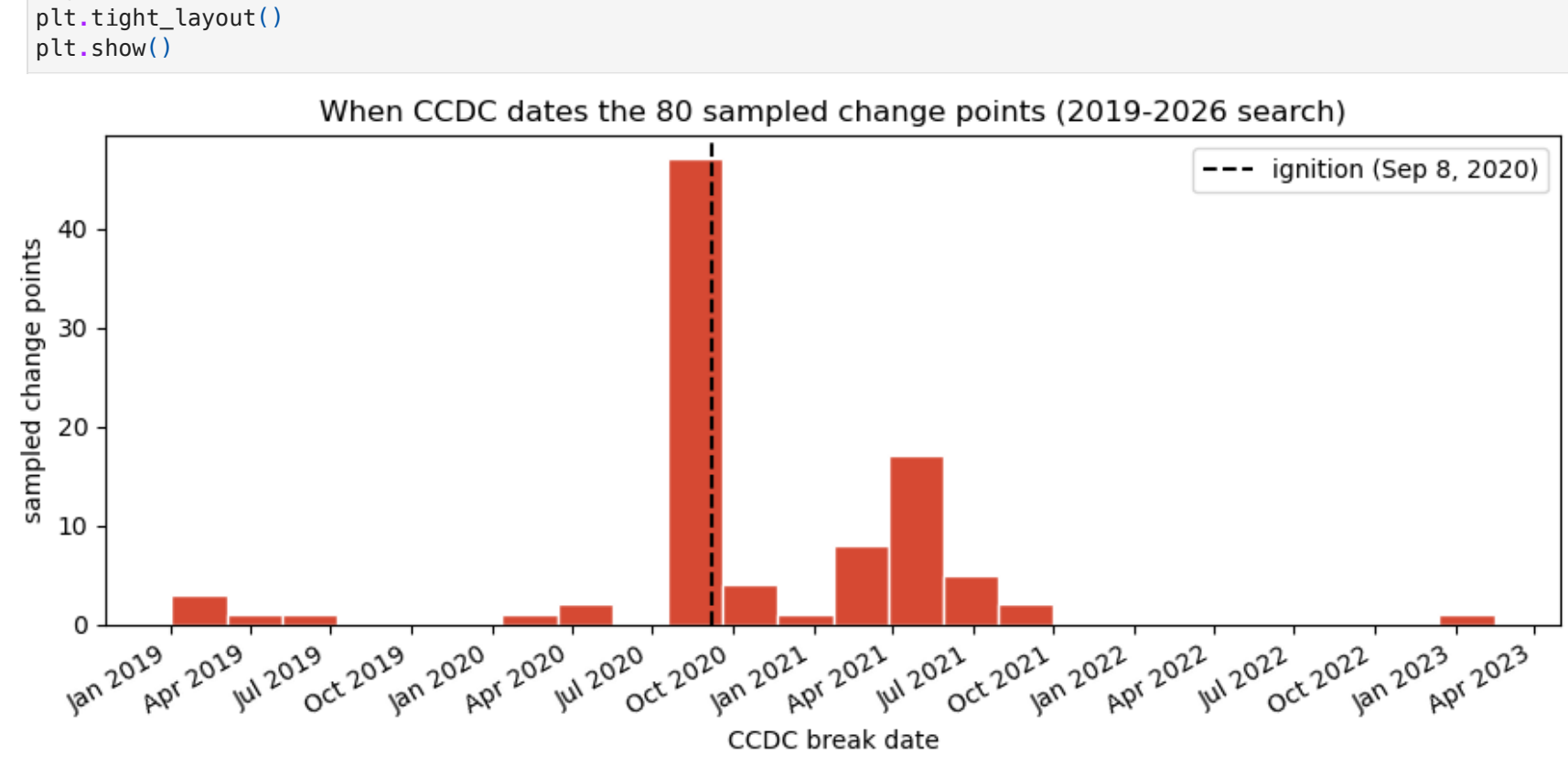
In [9]:

```
# ── Run (or load cached) CCDC breaks: 80 random + 10 water + 10 developed points ──────────
import time
import statistics as st
import rasterio.warp
from Change_Detection import prepare_landsat, run_ccdc, fire_window_break, frac_year_to_month

n_points = 80
n_water_target, n_developed_target = 10, 10
search_start_year, search_end_year = 2019, 2026  # searches for ANY break through end of 2025
# CCDC confirmation threshold. run_ccdc's default (6) needs 6 consecutive residual-
# exceeding observations to confirm a break; at noisier post-fire pixels that isn't met
# until the spring after a September fire, lagging the dated break ~6 months. Lowering to
# 4 lets the September discontinuity itself confirm the break. V.
ccdc_min_obs = 4
_tag = (f'{fire_name.title().replace(" ", "_")}_{search_start_year}_{search_end_year}'
        f'_n{n_points}_water{n_water_target}_dev{n_developed_target}')
ccdc_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_breaks_{_tag}.csv')
timing_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_timing_{_tag}.txt')

if rerun_CCDC or not os.path.exists(ccdc_path):
    # General random sample from the CLASSIFIER'S OWN Trees->non-Trees change map (tree_loss,
    # Step 6) -- not restricted to MTBS high severity -- so the check covers any severity and
    # any non-fire cause the classifier may have flagged.
    rng = np.random.default_rng(42)
    change_rows, change_cols = np.where(tree_loss)
    sel = rng.choice(len(change_rows), size=n_points, replace=False)
    sel_rows, sel_cols = change_rows[sel], change_cols[sel]
    sample_type = ['random'] * n_points

    # Stratified addition: explicitly include up to n_water_target Trees->Water and
    # n_developed_target Trees->Developed points, ADDED on top of the random 50 (not drawn
    # from it), so the general sample stays an unbiased draw. These transitions are under 1%
    # of the change area combined, so the random 50 almost never includes them by chance --
    # the first (50-only) run drew zero of either. Falls back to "as many as exist" if fewer
    # than the target are present in the change mask.
    for cls, target, label in [(1, n_water_target, 'stratified_water'),
                               (3, n_developed_target, 'stratified_developed')]:
        cls_rows, cls_cols = np.where(tree_loss & (a25 == cls))
        n_take = min(target, len(cls_rows))
        if n_take < target:
            print(f'Only {len(cls_rows)} Trees->{class1_dict[cls]} pixel(s) exist in the '
                  f'change mask -- sampling all {n_take} (wanted {target}).')
        if n_take > 0:
            take_idx = rng.choice(len(cls_rows), size=n_take, replace=False)
            sel_rows = np.concatenate([sel_rows, cls_rows[take_idx]])
            sel_cols = np.concatenate([sel_cols, cls_cols[take_idx]])
            sample_type += [label] * n_take

    n_total = len(sel_rows)

    # Pixel centers (EPSG:5070, from Step 6) -> lon/lat (EPSG:4326) for Earth Engine.
    xs, ys = rasterio.transform.xy(transform, sel_rows, sel_cols)
    lons, lats = rasterio.warp.transform(crs, 'EPSG:4326', xs, ys)
    to_classes = a25[sel_rows, sel_cols]  # each point's 2025 destination class, read locally

    points_fc = ee.FeatureCollection([
        ee.Feature(ee.Geometry.Point([lon, lat]), {'pid': i})
        for i, (lon, lat) in enumerate(zip(lons, lats))
    ])

    # MTBS severity at each point, ONE batched call -- a plain int band, not CCDC's expensive
    # array output, so (unlike CCDC itself) all points can be reduced together safely.
    # reduceRegions' output FeatureCollection is NOT guaranteed to preserve input order (an
    # earlier 60-point run's aggregate_array('first') "worked" at n=50 by coincidence, then
    # silently broke at n=60) -- key results by the 'pid' property explicitly instead.
    sev_fc_results = (get_severity_image(fire_year)
                      .reduceRegions(points_fc, ee.Reducer.first(), scale=30)
                      .getInfo()['features'])
    sev_by_pid = {f['properties']['pid']: f['properties'].get('first') for f in sev_fc_results}
    sev_vals = [sev_by_pid.get(i) for i in range(n_total)]

    # Landsat bracketing 2019-2025 (2 yr lead-in; trailing data through today for late-2025
    # breaks, though that tail is thinner than earlier events get, so those dates are less
    # confidently confirmed), then CCDC, then any break inside the search window.
    region = points_fc.geometry().buffer(90)  # ~3-pixel neighbourhoods; keeps CCDC tractable
    landsat = prepare_landsat(region, '2017-01-01', '2026-08-01')
    ccdc = run_ccdc(landsat, min_observations=ccdc_min_obs)
    break_img = fire_window_break(ccdc, search_start_year, search_end_year)

    # Sample one point at a time -- reducing all at once blows the memory limit. Timed, and
    # the elapsed time saved alongside the results (below).
    start_time = time.perf_counter()
    rows = []
    for i, (lon, lat) in enumerate(zip(lons, lats)):
        v = break_img.reduceRegion(ee.Reducer.first(), ee.Geometry.Point([lon, lat]), 30).get('break_year').getInfo()
        sv = sev_vals[i]
        rows.append({
            'lon': lon, 'lat': lat,
            'break_year': v if v and v > 0 else None,
            'severity': sv,
            'severity_name': SEVERITY_NAMES.get(int(sv), 'unknown') if sv is not None else 'outside perimeter',
            'to_class': int(to_classes[i]),
            'to_class_name': class1_dict.get(int(to_classes[i]), str(to_classes[i])),
            'sample_type': sample_type[i],
        })
    elapsed_sec = time.perf_counter() - start_time
    ccdc_breaks = pd.DataFrame(rows)

    os.makedirs(CCDC_OUTPUTS_DIR, exist_ok=True)
    ccdc_breaks.to_csv(ccdc_path, index=False)
    with open(timing_path, 'w') as f:
        f.write(f'{elapsed_sec:.2f} seconds for {n_total} points ({elapsed_sec / n_total:.2f} s/point)\n')
    print(f'Ran CCDC fresh -- saved {len(ccdc_breaks)} points -> {ccdc_path}')
    print(f'CCDC point-loop time: {elapsed_sec:.2f} s ({elapsed_sec / n_total:.2f} s/point) -> {timing_path}')
else:
    ccdc_breaks = pd.read_csv(ccdc_path)
    with open(timing_path) as f:
        print(f'Cached run took: {f.read().strip()}')
    print(f'Loaded cached CCDC breaks ({len(ccdc_breaks)} points) <- {ccdc_path}')
    print('(Set rerun_CCDC = True in the Setup cell to recompute from scratch.)')

breaks = ccdc_breaks['break_year'].dropna().tolist()

print(f'\nCCDC break in [{search_start_year}, {search_end_year}) at '
      f'{len(breaks)}/{len(ccdc_breaks)} sampled points')
if breaks:
    med = st.median(breaks)
    print(f'median break date : {med:.3f} (fractional years)  ->  month {frac_year_to_month(med):.1f}')
    print(f'range             : {min(breaks):.3f} .. {max(breaks):.3f}')
    print(f'fire ignition ref : 2020-09-08 = 2020.688  ->  month 9.3')

print('\nby sample type (breaks found / sampled):')
print(ccdc_breaks.groupby('sample_type')['break_year'].apply(lambda s: f'{s.notna().sum()}/{len(s)}').to_string())
print('\nby severity (breaks found / sampled, random points only):')
print(ccdc_breaks.loc[ccdc_breaks['sample_type'] == 'random']
      .groupby('severity_name')['break_year'].apply(lambda s: f'{s.notna().sum()}/{len(s)}').to_string())
```

```
Ran CCDC fresh -- saved 100 points -> ./CCDC_Outputs/ccdc_breaks_Holiday_Farm_2019_2026_n80_water10_dev10.csv
CCDC point-loop time: 733.34 s (7.33 s/point) -> ./CCDC_Outputs/ccdc_timing_Holiday_Farm_2019_2026_n80_water10_dev10.txt

CCDC break in [2019, 2026) at 93/100 sampled points
median break date : 2020.691 (fractional years)  ->  month 9.3
range             : 2019.004 .. 2023.122
fire ignition ref : 2020-09-08 = 2020.688  ->  month 9.3

by sample type (breaks found / sampled):
sample_type
random                  80/80
stratified_developed     9/10
stratified_water         4/10

by severity (breaks found / sampled, random points only):
severity_name
high        45/45
low           7/7
moderate    28/28
```

In [10]:

```
# ── Timeline of break dates (real calendar dates -- breaks now span multiple years) ─────────
import datetime as dt
import matplotlib.dates as mdates
import matplotlib.pyplot as plt
from Change_Detection import frac_year_to_date
rerun_CCDC = True
break_dates = [frac_year_to_date(b) for b in breaks]
fig, ax = plt.subplots(figsize=(9, 4))
ax.hist(break_dates, bins=24, color='#d64933', edgecolor='white')
ax.axvline(mdates.date2num(dt.date(2020, 9, 8)), color='k', ls='--', label='ignition (Sep 8, 2020)')
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
ax.set_xlabel('CCDC break date')
ax.set_ylabel('sampled change points')
ax.set_title(f'When CCDC dates the {n_points} sampled change points ({search_start_year}-{search_end_year} search)')
ax.legend()
fig.autofmt_xdate()
plt.tight_layout()
plt.show()
```

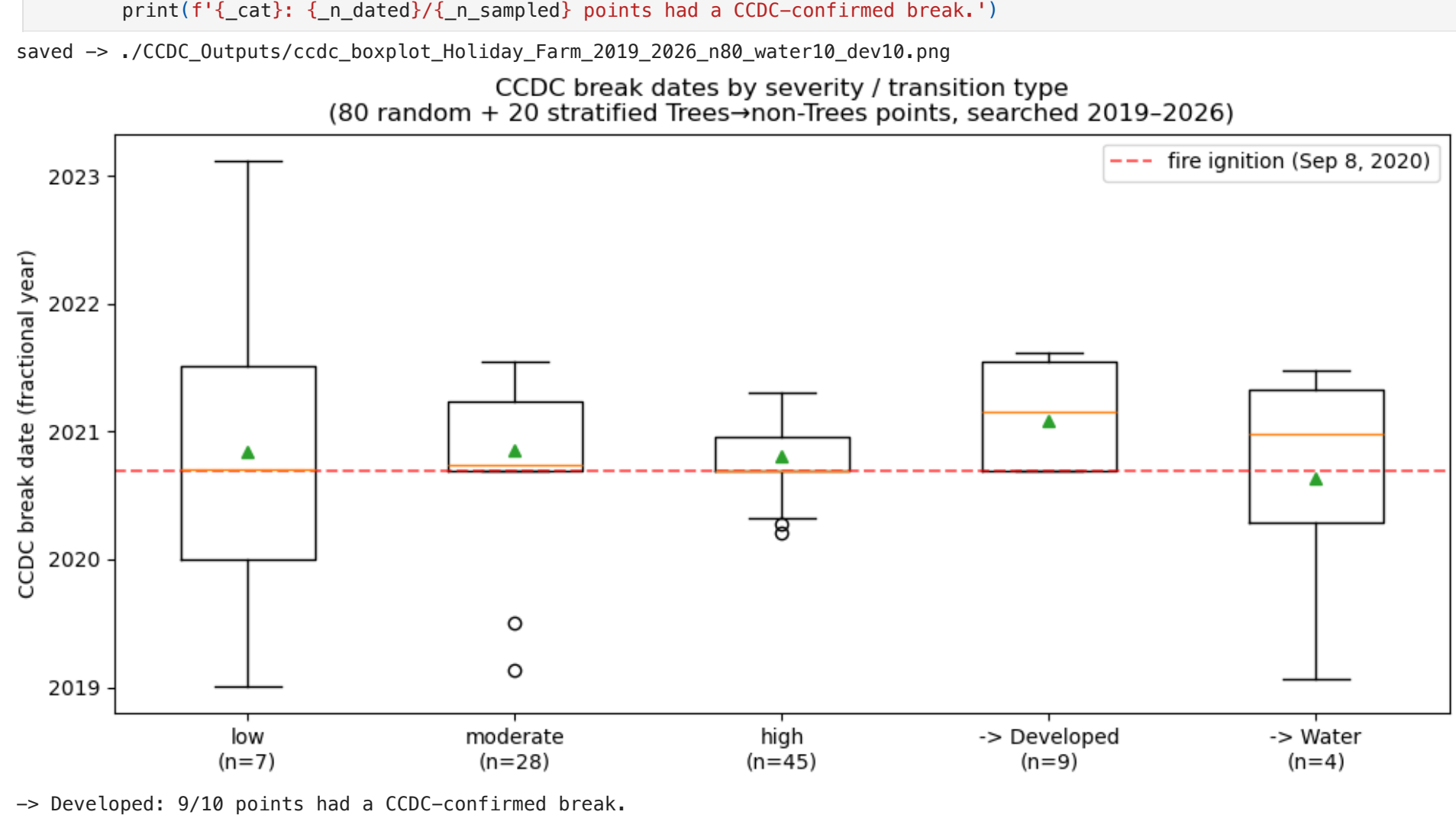


When CCDC dates the 80 sampled change points (2019-2026 search)

In [11]:

```
# ── Boxplot: break date by burn severity, and by Trees->Developed / Trees->Water ───────────
import matplotlib.pyplot as plt
# Category = the transition override (Developed/Water) if the point converted to one of those
# rare classes, else its MTBS severity. Developed/Water are always shown, even with 0 points --
# an empty box with "(n=0)" is the honest result for a rare-class random draw, not a bug to hide.
def _category(row):
    if row['to_class_name'] in ('Developed', 'Water'):
        return f"-> {row['to_class_name']}"
    return row['severity_name']

ccdc_breaks['category'] = ccdc_breaks.apply(_category, axis=1)

_severity_order = ['unburned/low', 'low', 'moderate', 'high', 'increased greenness', 'non-mapping']
_present = set(ccdc_breaks['category'])
_order = [c for c in _severity_order if c in _present] + ['-> Developed', '-> Water']

groups, tick_labels = [], []
for cat in _order:
    vals = ccdc_breaks.loc[ccdc_breaks['category'] == cat, 'break_year'].dropna().tolist()
    groups.append(vals)
    tick_labels.append(f'{cat}\n(n={len(vals)})')

fig, ax = plt.subplots(figsize=(10, 5))
ax.boxplot(groups, tick_labels=tick_labels, showmeans=True)
ax.axhline(2020.688, color='r', ls='--', alpha=0.6, label='fire ignition (Sep 8, 2020)')
ax.set_ylabel('CCDC break date (fractional year)')
ax.set_title(f'CCDC break dates by severity / transition type\n'
             f'({n_points} random + {len(ccdc_breaks) - n_points} stratified Trees\u2192non-Trees points, '
             f'searched {search_start_year}\u2013{search_end_year})')
ax.legend()
plt.tight_layout()

_boxplot_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_boxplot_{_tag}.png')
plt.savefig(_boxplot_path, dpi=150, bbox_inches='tight')
print(f'saved -> {_boxplot_path}')
plt.show()

print()
for _cat in ['-> Developed', '-> Water']:
    _sub = ccdc_breaks[ccdc_breaks['category'] == _cat]
    _n_sampled, _n_dated = len(_sub), _sub['break_year'].notna().sum()
    if _n_sampled == 0:
        print(f'{_cat}: no such points existed in the change mask.')
    elif _n_dated == 0:
        print(f'{_cat}: {_n_sampled} points sampled, CCDC found a break at NONE of them -- '
              f'check their MTBS severity above; no break + no fire severity usually means '
              f'classifier noise, not real change.')
    else:
        print(f'{_cat}: {_n_dated}/{_n_sampled} points had a CCDC-confirmed break.')
```

```
saved -> ./CCDC_Outputs/ccdc_boxplot_Holiday_Farm_2019_2026_n80_water10_dev10.png
```



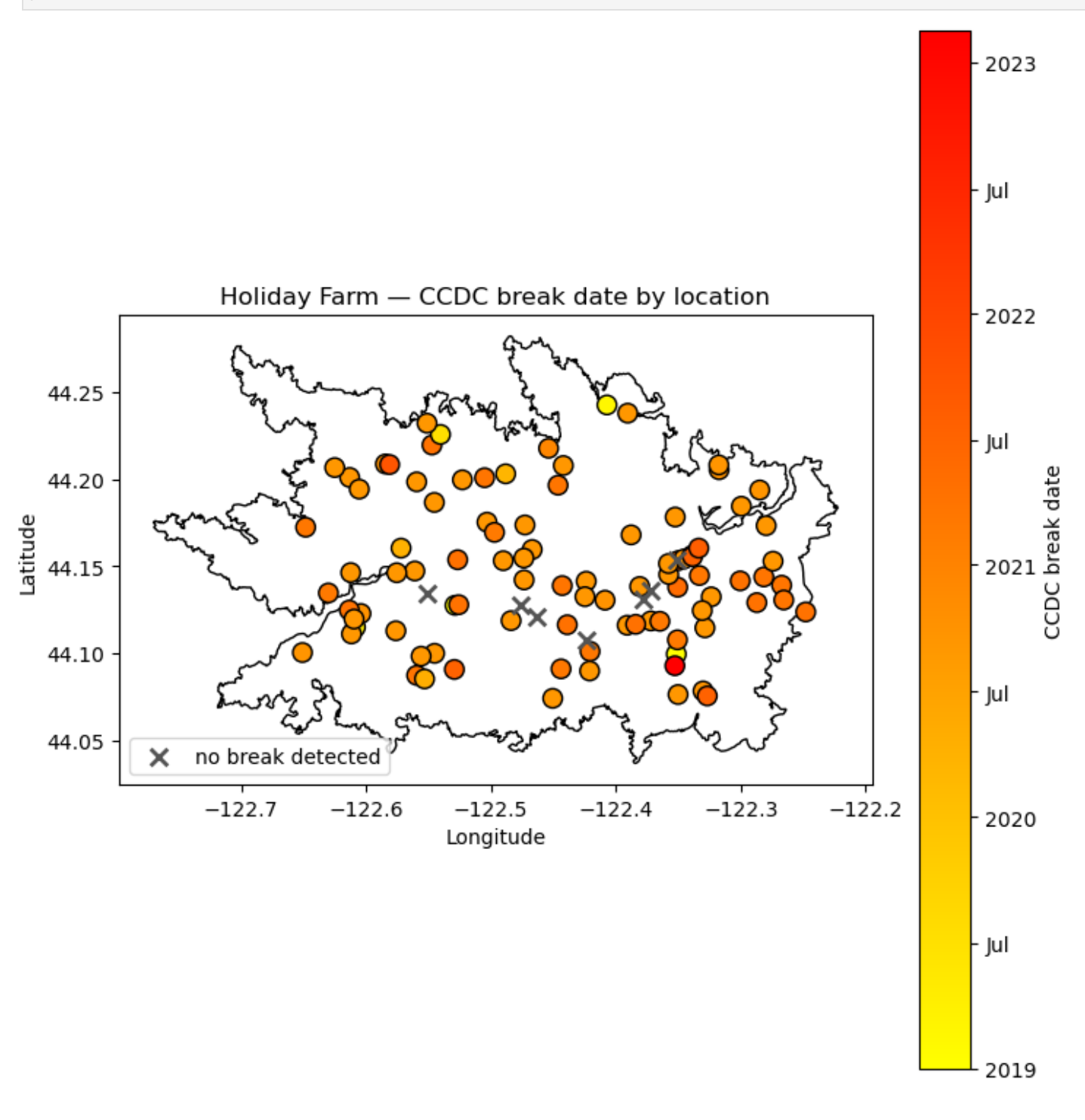
CCDC break dates by severity / transition type
(80 random + 20 stratified Trees→non-Trees points, searched 2019–2026)

```
-> Developed: 9/10 points had a CCDC-confirmed break.
-> Water: 4/10 points had a CCDC-confirmed break.
```

The central tendency and spread for the different groups are consistent with expectations. The three areas that burned all have central tendencies around the fire ignition date with the low burn having a wider spread. The wider spread suggests that the land cover change occurred from more than just the Labor Day fires. The developed pixels have a central tendency later than the fire ignition date, which suggests that the deforestation to developed is reasonable and occurred separate from the fire.

In [12]:

```
# ── Map: CCDC break date by location (color = when each point burned) ─────────────────────
import geopandas as gpd
import matplotlib.dates as mdates
import matplotlib.pyplot as plt
from shapely.geometry import shape
from Change_Detection import frac_year_to_date

ccdc_breaks['break_date'] = ccdc_breaks['break_year'].apply(
    lambda v: frac_year_to_date(v) if pd.notna(v) else pd.NaT)
has_break = ccdc_breaks['break_date'].notna()

fire_shape = gpd.GeoDataFrame(geometry=[shape(fire_geometry.getInfo())], crs='EPSG:4326')

# Autoscale color to the OBSERVED date range, not the full year -- with breaks clustered in a
# ~3-week window, a Jan-Dec scale would make every point look nearly identical.
dates_num = mdates.date2num(ccdc_breaks.loc[has_break, 'break_date'])
vmin, vmax = dates_num.min() - 2, dates_num.max() + 2

fig, ax = plt.subplots(figsize=(7.5, 7.5))
fire_shape.boundary.plot(ax=ax, color='black', linewidth=1)
sc = ax.scatter(ccdc_breaks.loc[has_break, 'lon'], ccdc_breaks.loc[has_break, 'lat'],
                c=dates_num, cmap='autumn_r', s=90, edgecolor='black', vmin=vmin, vmax=vmax, zorder=3)
ax.scatter(ccdc_breaks.loc[~has_break, 'lon'], ccdc_breaks.loc[~has_break, 'lat'],
           marker='x', color='#555555', s=70, linewidths=2, label='no break detected', zorder=3)
cbar = fig.colorbar(sc, ax=ax)
cbar.set_label('CCDC break date')
cbar.ax.yaxis.set_major_formatter(mdates.ConciseDateFormatter(mdates.AutoDateLocator()))
cbar.ax.yaxis.set_major_locator(mdates.AutoDateLocator())
ax.set_title(f'{fire_name.title()} — CCDC break date by location')
ax.set_xlabel('Longitude'); ax.set_ylabel('Latitude')
ax.legend(loc='lower left')
plt.tight_layout()
plt.show()
```



Holiday Farm — CCDC break date by location

NDVI / NBR time series at every sampled point

One diagnostic plot per point: NDVI and NBR (burn index) over 2017-2026, with the fire ignition date and (if found) the CCDC break date both marked. This is what actually justifies trusting or distrusting a CCDC date, rather than taking the fitted break on faith.

A concrete example worth opening directly

The points CCDC dated to *before* the fire (search the printed list below for `break_year < 2020`) show a genuine, permanent step-down in both indices around their break date, not a crash at the fire's September 2020 ignition. That is independent visual confirmation those are real, separate (likely harvest) disturbances, not a CCDC dating artifact. Compare against a random high-severity point, where both indices instead crash sharply right at the fire date and recover slowly over the following years (the textbook post-fire signature), and against a stratified Water point with no CCDC break, where neither index shows any discontinuity anywhere in the full record, corroborating the classifier-noise finding above with an independent signal.

Saved to `CCDC_Outputs/timeseries_{tag}/`, one PNG per point (`pt{{index}}_{{sample_type}}_{{break|nobreak}}.png`); a handful are shown inline below.

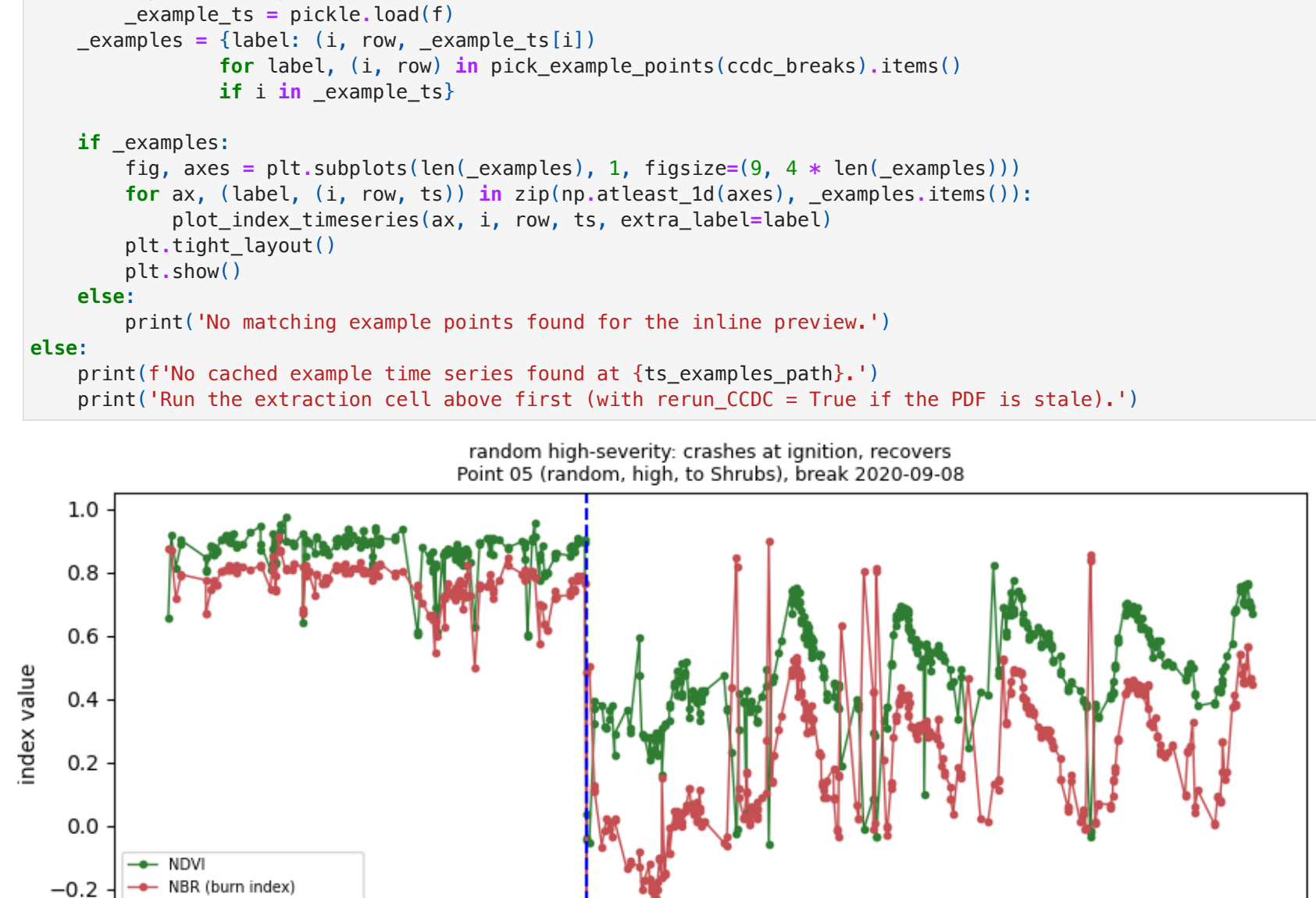
In [13]:

```
# ── NDVI / NBR time series for every sampled point, saved as one PDF, break date marked ────
import pickle
import time as _time
from matplotlib.backends.backend_pdf import PdfPages
from Change_Detection import (prepare_landsat, add_indices, point_index_timeseries,
                              plot_index_timeseries, pick_example_points)

ts_pdf_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_timeseries_{_tag}.pdf')
ts_timing_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_timeseries_timing_{_tag}.txt')
ts_examples_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_timeseries_examples_{_tag}.pkl')

if rerun_CCDC or not os.path.exists(ts_pdf_path):
    # Standalone (works whether Step 7 just ran fresh or loaded from cache): rebuilds the
    # Landsat collection from ccdc_breaks' own lon/lat rather than depending on the Run/load
    # cell's in-session `landsat`/`region`.
    _region = ee.Geometry.MultiPoint(ccdc_breaks[['lon', 'lat']].values.tolist()).buffer(90)
    _idx_collection = prepare_landsat(_region, '2017-01-01', '2026-08-01').map(add_indices)
    _example_indices = {i for i, _ in pick_example_points(ccdc_breaks).values()}

    print(f'Extracting NDVI/NBR time series for {len(ccdc_breaks)} points '
          f'(~{7 * len(ccdc_breaks) / 60:.0f}s expected)...')
    _start = _time.perf_counter()
    _example_ts = {}
    with PdfPages(ts_pdf_path) as _pdf:
        for i, row in ccdc_breaks.iterrows():
            ts = point_index_timeseries(_idx_collection, row['lon'], row['lat'])

            fig, ax = plt.subplots(figsize=(9, 4))
            plot_index_timeseries(ax, i, row, ts)
            plt.tight_layout()
            _pdf.savefig(fig)
            plt.close(fig)

            if i in _example_indices:
                _example_ts[i] = ts  # kept for the standalone preview cell below
    _elapsed = _time.perf_counter() - _start

    with open(ts_timing_path, 'w') as f:
        f.write(f'{_elapsed:.2f} seconds for {len(ccdc_breaks)} points ({_elapsed / len(ccdc_breaks):.2f} s/point)\n')
    with open(ts_examples_path, 'wb') as f:
        pickle.dump(_example_ts, f)
    print(f'Saved {len(ccdc_breaks)} time series pages -> {ts_pdf_path}')
    print(f'Time-series extraction time: {_elapsed:.2f} s ({_elapsed / len(ccdc_breaks):.2f} s/point) -> {ts_timing_path}')
else:
    with open(ts_timing_path) as f:
        print(f'Cached run took: {f.read().strip()}')
    print(f'Loaded cached time-series PDF <- {ts_pdf_path}')
    print('(Set rerun_CCDC = True in the Setup cell to recompute from scratch.)')
```

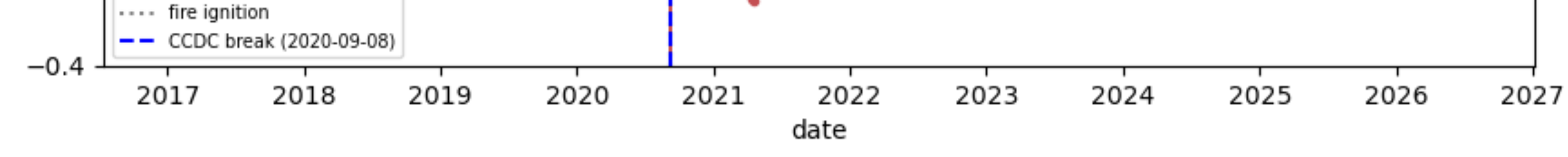
```
Extracting NDVI/NBR time series for 100 points (~12s expected)...
Saved 100 time series pages -> ./CCDC_Outputs/ccdc_timeseries_Holiday_Farm_2019_2026_n80_water10_dev10.pdf
Time-series extraction time: 803.47 s (8.03 s/point) -> ./CCDC_Outputs/ccdc_timeseries_timing_Holiday_Farm_2019_2026_n80_water10_dev10.txt
```

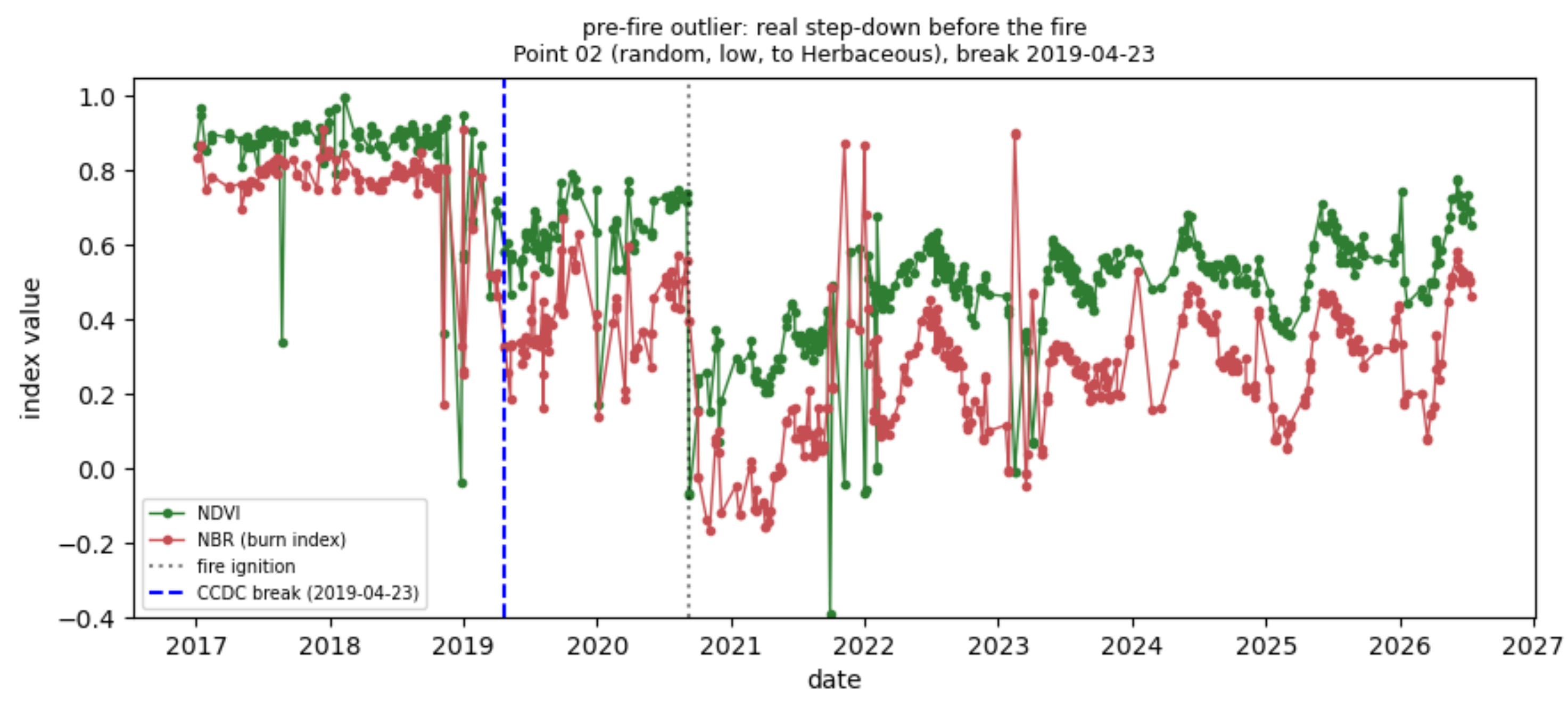
In [14]:

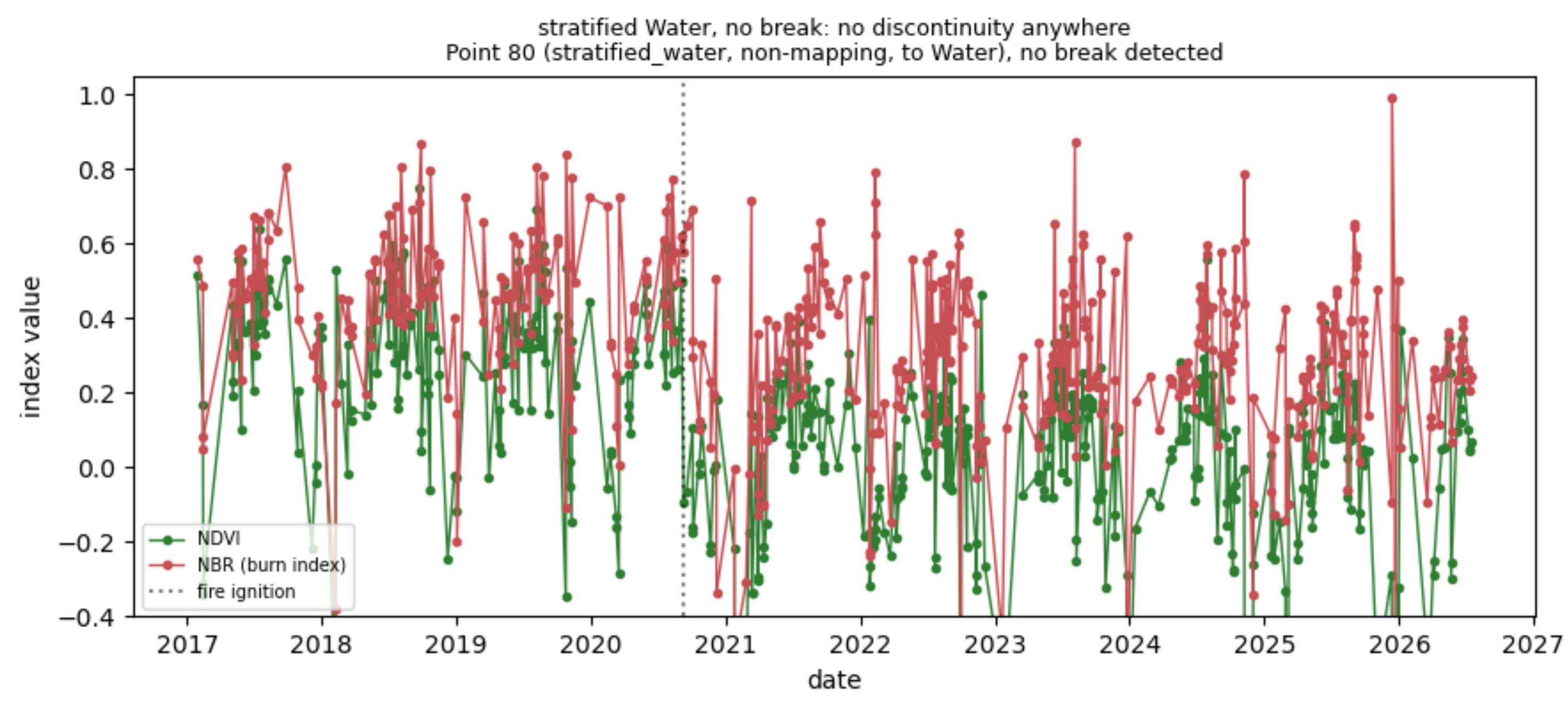
```
# ── Preview a few representative examples inline, read from disk (no GEE call needed) ──────
# Fully standalone: reads the example points' time series back from the small pickle saved
# alongside the PDF above, so this cell runs with rerun_CCDC = False even in a fresh kernel
# that never ran the extraction cell this session -- no Earth Engine query either way.
import pickle
from Change_Detection import plot_index_timeseries, pick_example_points

ts_examples_path = os.path.join(CCDC_OUTPUTS_DIR, f'ccdc_timeseries_examples_{_tag}.pkl')

if os.path.exists(ts_examples_path):
    with open(ts_examples_path, 'rb') as f:
        _example_ts = pickle.load(f)
    _examples = {label: (i, row, _example_ts[i])
                 for label, (i, row) in pick_example_points(ccdc_breaks).items()
                 if i in _example_ts}

    if _examples:
        fig, axes = plt.subplots(len(_examples), 1, figsize=(9, 4 * len(_examples)))
        for ax, (label, (i, row, ts)) in zip(np.atleast_1d(axes), _examples.items()):
            plot_index_timeseries(ax, i, row, ts, extra_label=label)
        plt.tight_layout()
        plt.show()
    else:
        print('No matching example points found for the inline preview.')
else:
    print(f'No cached example time series found at {ts_examples_path}.')
    print('Run the extraction cell above first (with rerun_CCDC = True if the PDF is stale).')
```

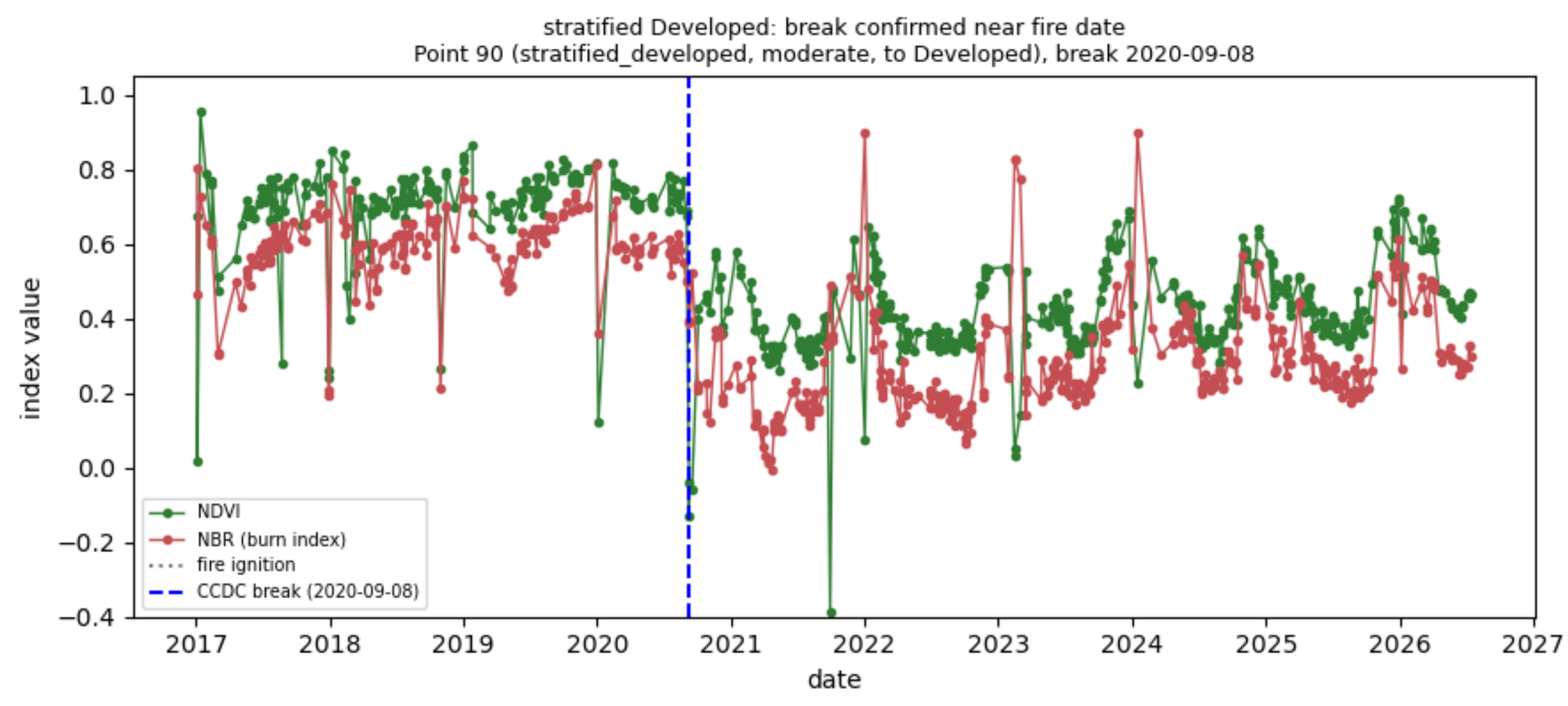


random high-severity: crashes at ignition, recovers
Point 05 (random, high, to Shrubs), break 2020-09-08




pre-fire outlier: real step-down before the fire
Point 02 (random, low, to Herbaceous), break 2019-04-23


stratified Water, no break: no discontinuity anywhere
Point 80 (stratified_water, non-mapping, to Water), no break detected


stratified Developed: break confirmed near fire date
Point 90 (stratified_developed, moderate, to Developed), break 2020-09-08

Step 8: Cross-check the date against Hansen

CCDC (Step 7) is our own dating; Hansen Global Forest Change is an independent, pre-computed forest-loss product (`UMD/hansen/global_forest_change_2025_v1_13` , `lossyear` band, ~30 m annual). If both say 2020, the date is corroborated by a source with entirely different methods and inputs.

Hansen is annual (coarser than CCDC's sub-annual break), and it has a known quirk: late-season fires spill partly into the *next* year, because the loss is only confirmed in the following year's cloud-free imagery. So a September-2020 burn shows up as mostly 2020 with a chunk of 2021; both are the fire.

In [15]:
```
# ── Hansen loss-year distribution within the MTBS high-severity zone ──────────────
from Change_Detection import loss_year_table

high_sev = get_severity_image(fire_year).eq(4).selfMask()
hansen_tbl = loss_year_table(fire_geometry, mask=high_sev)

print('Hansen forest-loss year within the high-severity zone:')
display(hansen_tbl.sort_values('hectares', ascending=False).head(6).round(1))

fire_frac = hansen_tbl.reindex([fire_year, fire_year + 1])['fraction'].sum()
print(f'Any loss detected     : {hansen_tbl["fraction"].sum() * 100:5.1f}% of the high-severity zone')
print(f'Dated {fire_year}              : {hansen_tbl["fraction"].get(fire_year, 0) * 100:5.1f}%')
print(f'Dated {fire_year} or {fire_year + 1}       : {fire_frac * 100:5.1f}%   (Hansen lags late-season fires ~1 yr)')
print(f'\nAgreement: CCDC (Sep 8, {fire_year}), Hansen ({fire_year}/{fire_year + 1}), and MTBS '
      f'(ignited {fire_year}-09-08) all place the change in {fire_year}.')
```

```
Hansen forest-loss year within the high-severity zone:
```

year	hectares	fraction
2020	16234.1	0.7
2021	5860.8	0.3
2019	181.7	0.0
2002	179.3	0.0
2003	171.4	0.0
2001	167.8	0.0

```
Any loss detected     : 100.0% of the high-severity zone
Dated 2020            :  69.4%
Dated 2020 or 2021       :  94.4%   (Hansen lags late-season fires ~1 yr)

Agreement: CCDC (Sep 8, 2020), Hansen (2020/2021), and MTBS (ignited 2020-09-08) all place the change in 2020.
```

Step 9: Activity data (area of change by county)

The first step of carbon accounting is activity data: *how much* forest was lost, per reporting domain (county). This attributes the Trees to non-Trees change to the counties the fire spans by rasterizing the county polygons onto the change grid, so each pixel lands in exactly one county and the split reconciles with the total. (The remaining carbon piece, multiplying by a per-county FIA emission factor, is Step 10, still to wire up.)

In [16]:
```
# ── Attribute the Trees->non-Trees change to counties ─────────────────────────────
from Geo_Utils import get_counties
from Carbon_Accounting import area_by_polygons

# The Holiday Farm Fire spans Lane and Linn (both Oregon). For the full WA+OR run, drop the
# `names` filter and key on GEOID -- county NAMEs collide across the two states.
PILOT_COUNTY_NAMES = ['Lane', 'Linn']
counties = get_counties(names=PILOT_COUNTY_NAMES)
county_area, outside_ha = area_by_polygons(tree_loss, transform, crs, counties, label_col='NAME')

print('Trees -> non-Trees area by county:')
display(county_area.round(0))
print(f'outside Lane/Linn  : {outside_ha:>8,.0f} ha   (should be ~0 -> fire fully inside these two)')
print(f'reconcile          : {county_area["hectares"].sum() + outside_ha:>8,.0f} ha   '
      f'vs {detected_ha:,.0f} ha detected')
```

```
Trees -> non-Trees area by county:
```

NAME	hectares	fraction
Lane	34203.0	1.0
Linn	1964.0	0.0

```
outside Lane/Linn  :        0 ha   (should be ~0 -> fire fully inside these two)
reconcile          :   36,167 ha   vs 36,167 ha detected
```

Step 10: Aboveground carbon lost (Activity Data × Emission Factor)

Carbon lost = area of forest lost (Step 9) × carbon density (Mg C/ha). This is the standard IPCC/VCS accounting form, and it's the reason the county split matters: carbon is reported as county totals, not a per-pixel map.

Emission factors come from FIA, not from a model

FIA gives design-based aboveground biomass per county with a sampling error (Tier 2, free), and its ~1-mile plot-coordinate fuzzing is irrelevant at county scale. Convert biomass to carbon with the IPCC 0.47 fraction (0.50 shown as a conifer sensitivity).

FIA is a manual pull

It is not in Earth Engine and its API is not reachable here, so the `FIA_BIOMASS_EF` values were manually pulled from FIA EVALIDator.

Verra framing

The AD × EF structure and Tier-2 FIA factors are conventional. But two gaps make this jurisdictional-style, not crediting-grade: (1) the area is map-counted, biased high and with no confidence interval (Olofsson sample-based estimation is the accepted fix); so (2) the ± below reflects FIA emission-factor uncertainty only and understates the total. No project-specific field plots, no conservativeness deductions, no VVB review. So these reported values are estimates only.

In [17]:
```
# ── Emission factors: FIA aboveground biomass (Mg/ha) per county ──────────────────
from Carbon_Accounting import carbon_from_area, convert_tons_acre_to_mg_ha

FIA_EF_df = pd.read_csv(f'{FIA_DATA_DIR}/FIADB_API_Estimates_183129161.csv')

FIA_BIOMASS_EF = {
}

for county in PILOT_COUNTY_NAMES:
    county_EF_df = FIA_EF_df.loc[FIA_EF_df['COUNTY_CODE_AND_NAME'].str.contains(county,na=False),]
    county_EF = county_EF_df.loc[county_EF_df['OWNERSHIP_GROUP'] == "Total",]
    county_AGB = convert_tons_acre_to_mg_ha(county_EF['RATIO_ESTIMATE'].iloc[0])
    county_AGB_SE = convert_tons_acre_to_mg_ha(county_EF['RATIO_SE'].iloc[0])
    FIA_BIOMASS_EF[county] = (county_AGB,county_AGB_SE)

# ── AD x EF, with the IPCC 0.47 carbon fraction and a 0.50 conifer sensitivity ────────
carbon = carbon_from_area(county_area, FIA_BIOMASS_EF, carbon_fraction=0.47)
carbon_hi = carbon_from_area(county_area, FIA_BIOMASS_EF, carbon_fraction=0.50)

print(f'Aboveground carbon lost, Holiday Farm Fire (Lane + Linn)')
display(carbon.round(0))

tot, se = carbon.loc['TOTAL', 'carbon_mg'], carbon.loc['TOTAL', 'carbon_se_mg']
print(f'\nTotal (x0.47): {tot:>14,.0f} +/- {se:,.0f} Mg C   (EF uncertainty only)')
print(f'Total (x0.50): {carbon_hi.loc["TOTAL","carbon_mg"]:>14,.0f} Mg C   (conifer sensitivity)')
print(f'Implied EF   : {tot / county_area["hectares"].sum():.0f} Mg C/ha   '
      f'(PNW conifer sanity range ~100-250)')
```

```
Aboveground carbon lost, Holiday Farm Fire (Lane + Linn)
```

NAME	hectares	agb_mg_ha	agb_se_mg_ha	carbon_mg	carbon_se_mg
Lane	34203.0	157.0	5.0	2525783.0	73944.0
Linn	1964.0	120.0	6.0	111082.0	5815.0
TOTAL	36167.0	NaN	NaN	2636866.0	74172.0

```
Total (x0.47):      2,636,866 +/- 74,172 Mg C   (EF uncertainty only)
Total (x0.50):      2,805,176 Mg C   (conifer sensitivity)
Implied EF   : 73 Mg C/ha   (PNW conifer sanity range ~100-250)
```

Conclusions

The end-to-end pipeline works and holds up against independent truth

Running the whole chain on a real, dated, well-measured fire (classify, difference the two maps, confirm and date the change with CCDC, attribute it to counties, and convert to carbon) produced results that line up with MTBS severity, Hansen loss year, and the known 2020-09-08 ignition. On the falsifiable predictions set out at the start, the pilot passes well enough to justify moving to the wall-to-wall Oregon run.

What the pilot got right

Area. Detected Trees to non-Trees change was 36,167 ha, sitting between the MTBS high-severity area (23,432 ha) and the high-plus-moderate area (44,214 ha), at 0.82 times the moderate-plus-high stratum. The map recovered the stand-replacing core plus some moderate-severity flip to shrubs, not the full 70,298 ha perimeter, which is the intended behavior and shows it is not grossly over-detecting.

Transitions are plausible. About 87% of the change was Trees to Shrubs, with the rest Herbaceous and Barren. The implausible transitions are correctly negligible: Trees to Water was 16 ha and Trees to Developed 202 ha, together under 1% of the change area.

Dating is accurate. CCDC found a break in the 2019 to 2025 window at all 80 random change points, and at 93 of 100 points overall, with a median break date of 2020-09-09, about one day after the 2020-09-08 ignition. Every high, moderate, and low severity random point registered a break, and the burned groups cluster tightly on the ignition date in the boxplot.

Cross-checks agree. Hansen reports loss across 100% of the high-severity core, 69% of it dated to 2020 and 94% to 2020 or 2021 (Hansen lags late-season fires by about a year). MTBS, Hansen, and CCDC all place the change in 2020.

County attribution is clean. The change splits 94% to Lane and 5% to Linn, with 0 ha outside the two counties and an exact reconciliation, so there is no border double-count.

CCDC works as the arbiter. Only 4 of 10 stratified Trees to Water points had a CCDC-confirmed break, consistent with those being classifier noise rather than real change. That is exactly the role CCDC plays: separating real change from map error before any carbon is attributed to it.

Caveats and what to keep in mind for the Oregon wall-to-wall run

Area is map-counted, biased high, and has no confidence interval. The 36,167 ha figure carries no error bar, and post-classification map counting is biased toward over-detection. Confirming change with CCDC removes some false positives, but we cannot say how much without a reference sample (the Olofsson good-practice estimator). This matters more at the Oregon scale, not less: the false-positive mass grows with the large stable area, so a state-wide change total should be read as an upper-leaning estimate until a reference-sample area correction is added.

The carbon emission factor is a county average that likely underestimates a mature-forest burn. The implied carbon density here is 73 Mg C per ha, below the 100 to 250 Mg C per ha range expected for mature PNW conifer. That is because the FIA county-mean aboveground biomass (157 Mg per ha for Lane) is diluted by young, managed, and recently harvested stands across the whole county, whereas this fire burned denser mature forest. So the pilot carbon total, about 2.6 million Mg C plus or minus 74,000 (EF uncertainty only), is plausible in order of magnitude but probably low for this specific event. Tightening it means forest-type or stand-age stratification, which is the concrete reason to bring in the Level-2 (Deciduous, Evergreen, Mixed) model and finer FIA strata for the full run. Note also that the reported interval is EF-only; the area term's bias is not priced in.

CCDC dating here is a best case. September is cloud-free in the PNW, so this validates the method but not winter (November to March) dating, which stays poorly constrained by Landsat density. The wider spread in the low-severity group and the scattering of 2019 and 2023 breaks reflect genuine non-fire change plus dating noise, both of which will be present across Oregon. A separate CCDC versus COLD comparison on these points (see COLD_vs_CCDC_Comparison) found COLD dates the fire even more tightly. COLD, however, cannot be run on GEE and would require money to run.

Scale and operations. The pilot covers about 700 km2; the Oregon forest AOI is about 171,000 km2, roughly 243 times larger. Expect the same qualitative behavior, but the noise and bias effects scale with area, and the operational pieces (VM runtime, auto-delete cutoffs, resumable inference) become load-bearing rather than incidental. The full model also has no Ice and Snow class, which the GLIMS glacier mask handles in the wall-to-wall AOI but was not needed for this low-elevation fire.

Bottom line

The methods work decently well. On a known event the pipeline recovers the right disturbance area to within the expected severity band, dates it to the correct month, attributes it to the correct counties, and produces a carbon total of the right order of magnitude, all cross-checked against independent data.